File

Name: ex8-Rz2.3
Description: TR30 example 8 ex8-Rz2.3 - UNACCEPTED
Comment: Original printed Rz=2.3; inherited candidate baseline. Native material validation and input identity are recorded separately.
Changed by: AAA on: 29.10.2026 at: 15:49:37

1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1 ----------	Gear 2 -------
Power (kW)	[P]	223.725	
Speed (1/min)	[n]	1430.0	340.5
Number of load cycles (in mio.)	[NL]	858.000	204.286
Torque (Nm)	[T]	1494.0	6274.8
Application factor	[KA]	1.00	
Required service life (h)	[H]	10000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to ISO 21771:2007 (withdrawn)

		------- Gear 1 ----------	Gear 2 -------
Center distance (mm)	[a]	182.880	
Center distance allowances (mm)	[Aa.e/i]	0.000	/ 0.000
Normal module (mm)	[mn]	4.5000	
Transverse module (mm)	[mt]	4.5842	
Normal diametral pitch (1/in)	[Pnd]	5.64444	
Transverse diametral pitch (1/in)	[Ptd]	5.54074	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	11.0000	
Number of teeth	[z]	15	63
Facewidth (mm)	[b]	78.74	76.20
Hand of gear		left	right
Accuracy grade	[Q-ISO 1328:2013]	A6	A6
Inner diameter (mm)	[di]	0.00	0.00
Shaft diameter (mm)	[ds]	77.58	377.50
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1 ----------	Gear 2 -------

3 Materials

Gear 1

Own input, TR30 example 8 Eh MQ, Case hardening steel, case-hardened, Original TR30 table properties; material limits from ISO 6336-5 Table 1

Gear 2

Own input, TR30 example 8 Eh MQ, Case hardening steel, case-hardened, Original TR30 table properties; material limits from ISO 6336-5 Table 1

		------- Gear 1	---------- Gear 2 -------
Surface hardness		HRC 58	HRC 58
Optimal quality and experience: Life factors Z_{NT} and $Y_{NT} \geq 1$			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	500.00	500.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	689.50	689.50

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.38	0.38
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	2.30	2.30
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	18.00	18.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Oil type, own input	TR30 example 8: nu40 input; other oil properties are ancillary defaults	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	220.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	19.00
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	80.000

4 Geometry

4.1 Reference profiles

4.1.1 Information on pre-machining

Reference profile Gear 1

Reference profile, own input	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Min. finishing stock (mm)	[qmin]	0.220
Protuberance (mm)	[pr0]	0.267
Remaining undercut (mm)	[spr0 = pr0-q]	0.047
Dedendum coefficient	[hfP*]	1.516
Root radius factor	[ρfP*]	0.400
	[ρfPmax*]	0.420
Addendum coefficient	[haP*]	1.021
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.633
Protuberance angle	[αprP]	10.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	37.461
(mm)	[a0.e/i]	37.215 /37.215
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	1.931 / 1.931

Reference profile Gear 2

Reference profile, own input	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	

Min. finishing stock (mm)	[qmin]	0.220
Protuberance (mm)	[pr0]	0.267
Remaining undercut (mm)	[spr0 = pr0-q]	0.047
Dedendum coefficient	[hfP*]	1.516
Root radius factor	[ρfP*]	0.400
	[ρfPmax*]	0.420
Addendum coefficient	[haP*]	1.047
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.633
Protuberance angle	[αprP]	10.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	147.023
(mm)	[a0.e/i]	146.877 /146.877
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	1.844 / 1.844

4.1.2 Reference profile of the final tooth form

		------- Gear 1	---------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.373	1.373
Tooth root radius reference profile	[ρfP*]	0.400	0.400
Addendum reference profile	[haP*]	1.021	1.047
Protuberance height coefficient	[hprP*]	0.355	0.355
Protuberance angle (°)	[αprP]	10.000	10.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Data for final machining:			
Depth of immersion	[hgrind*]	0.954	1.040
Radius at cutter head	[rgrind*]	0.100	0.100
Grinding only flank (0), flank & root (1)		0	0
Generation grinding (0), form grinding (1)		0	0
Type of profile modification:	none (only running-in)		
Tip relief by running in (µm)	[Ca L/R]	13.0 /13.0	13.0 /13.0

4.2 Basic data

Overall transmission ratio	[itot]	-4.200
Gear ratio	[u]	4.200
Transverse module (mm)	[mt]	4.584
Transverse pressure angle (°)	[αt]	20.344
Working pressure angle (°)	[αwt]	23.562
(°)	[αwt.e/i]	23.562 /23.562
Working pressure angle at normal section (°)	[αwn]	23.158
Base helix angle (°)	[βb]	10.329
Helix angle at operating pitch circle (°)	[βw]	11.246
Reference center distance (mm)	[ad]	178.785
Sum of profile shift coefficients	[Σxi]	0.9806

		------- Gear 1	---------- Gear 2 -------
Profile shift coefficient			
Information on pre-machining	[xV]	0.6842	0.5822
Information on final machining	[x]	0.5413	0.4393
Generating profile shift coefficient			
Information on pre-machining	[xEV.e/i]	0.6295/ 0.6295	0.5499/ 0.5499
Information on final machining	[xE.e/i]	0.4866/ 0.4866	0.4069/ 0.4069
Virtual gear no. of teeth	[zn]	15.788	66.311

Involute length (mm)		[l_dFa-l_dFf]	9.274	9.352

4.3 Diameters and their allowances

			------- Gear 1 ----------	Gear 2 -------
Reference diameter (mm)		[d]	68.763	288.806
Base diameter (mm)		[db]	64.474	270.791
Tip alteration (mm)		[k*mn]	-0.317	-0.317
Tip diameter (mm)		[da]	82.190	301.550
(mm)		[da.e/i]	82.190 /82.190	301.550 /301.550
Tip diameter allowances (mm)		[Ada.e/i]	0.000 / 0.000	0.000 / 0.000
Tip form diameter (mm)		[dFa]	82.190	301.550
(mm)		[dFa.e/i]	82.190 /82.190	301.550 /301.550
Root form diameter, pre-machining (mm)		[dFa_p]	82.190	301.550
	(mm)	[dFa_p.e/i]	82.190 /82.190	301.550 /301.550
Active tip diameter (mm)		[dNa]	82.190	301.550
(mm)		[dNa.e/i]	82.190 /82.190	301.550 /301.550
V-Circle diameter (mm)		[dv]	73.635	292.760
(mm)		[dv.e/i]	73.143 /73.143	292.469 /292.469
Operating pitch diameter (mm)		[dw]	70.338	295.422
(mm)		[dw.e/i]	70.338 /70.338	295.422 /295.422
Root diameter (mm)		[df]	61.278	280.402
(mm)		[df.e/i]	60.785 /60.785	280.111 /280.111
(calculated with pre-machining tool)				
Active root diameter (mm)		[dNf]	65.878	287.051
(mm)		[dNf.e/i]	65.878 /65.878	287.051 /287.051
Root form diameter (mm)		[dFf]	66.057	284.259
(mm)		[dFf.e/i]	65.875 /65.875	284.033 /284.033
(calculated with final machining tool)				
Root form diameter, pre-machining (mm)		[dFf_p]	67.539	286.295
	(mm)	[dFf_p.e/i]	67.370 /67.370	286.050 /286.050

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	1.584	1.466
Effective tip clearance (mm)	[c.e/i]	1.730 / 1.730	1.712 / 1.712
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.002 / 0.002	1.509 / 1.509
Undercut taken into account			
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	6.713	6.372
(mm)	[ha.e/i]	6.713 / 6.713	6.372 / 6.372
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	3.743	4.202
(mm)	[hf.e/i]	3.989 / 3.989	4.348 / 4.348
Tooth height (mm)	[h]	10.456	10.574

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	45.298 /45.298	28.074 /28.074
Roll angle to dNf (°)	[ξNf.e/i]	12.022 /12.022	20.151 /20.151
Roll angle at dFf (°)	[ξFf.e/i]	12.008 /12.008	18.136 /18.136

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.9648	1.8906
Normal tooth thickness at tip circle (mm)	[san]	2.008	3.352
(mm)	[san.e/i]	1.796 / 1.796	3.242 / 3.242
Normal tooth thickness at tip form circle (mm)	[sFan]	2.008	3.352
(mm)	[sFan.e/i]	1.796 / 1.796	3.242 / 3.242
Normal space width at root circle (mm)	[efn]	0.000	2.851
(mm)	[efn.e/i]	0.000 / 0.000	2.875 / 2.875
Pitch on reference circle (mm)	[pt]	14.402	14.402
Base pitch (mm)	[pbt]	13.503	13.503
Transverse pitch on contact-path (mm)	[pet]	13.503	13.503
Lead height (mm)	[pz]	1111.360	4667.712
Axial pitch (mm)	[px]	74.091	74.091

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	2.119	1.352
Specific sliding at the tip	[ζa]	0.555	0.572
Specific sliding at the root	[ζf]	-1.335	-1.248
Mean specific sliding	[ζm]	0.562	
Sliding factor on tip	[Kga]	0.402	0.257
Sliding factor on root	[Kgf]	-0.257	-0.402

4.8 Contact ratios

		---------- Pair ----------
Minimal length of contact line (mm)	[Lmin]	106.562
Transverse contact ratio	[εα]	1.386
	[εα.e/m/i]	1.386 / 1.386/ 1.386
Overlap ratio	[εβ]	1.028
Total contact ratio	[εγ]	2.415
	[εγ.e/m/i]	2.415 / 2.415/ 2.415
Length of path of contact (mm)	[gα]	18.722
(mm)	[gα.e/m/i]	18.722 /18.722/18.722
Approach path of contact (mm)	[gf]	7.294
(mm)	[gf.e/m/i]	7.294 / 7.294/ 7.294
Recess path of contact (mm)	[ga]	11.428
(mm)	[ga.e/m/i]	11.428 /11.428/11.428

		------- Gear 1 ----------	Gear 2 -------
Addendum contact ratio	[ε]	0.846	0.540
	[ε.e/i]	0.846 / 0.846	0.540 / 0.540
Length T1-A and T2-A (mm)	[T1A,T2A]	6.764	66.341
(mm)	[.e/i]	6.764 / 6.764	66.341 / 66.341

Length T1-B and T2-B (mm)	[T1B,T2B]	11.983	61.122
(mm)	[.e/i]	11.983 / 11.983	61.122 / 61.122
Length T1-C and T2-C (mm)	[T1C,T2C]	14.059	59.047
(mm)	[.e/i]	14.059 / 14.059	59.047 / 59.047
Length T1-D and T2-D (mm)	[T1D,T2D]	20.268	52.838
(mm)	[.e/i]	20.268 / 20.268	52.838 / 52.838
Length T1-E and T2-E (mm)	[T1E,T2E]	25.487	47.619
(mm)	[.e/i]	25.487 / 25.487	47.619 / 47.619
Length T1-T2 (mm)	[T1T2]	73.105	
(mm)	[.e/i]	73.105 / 73.105	

Diameter of single contact point B (mm)	[d-B]	68.784	297.105
(mm)	[d-B.e/i]	68.784 / 68.784	297.105 /297.105
Diameter of single contact point D (mm)	[d-D]	76.158	290.680
(mm)	[d-D.e/i]	76.158 / 76.158	290.680 /290.680

5 General influence factors

5.1 Forces and circumferential speed

		------- Gear 1 ----------	Gear 2 -------
Nominal circum. force at pitch circle (N)	[Ft]	43453.36	
Axial force (N)	[Fa]	8446.48	8446.48
Radial force (N)	[Fr]	16111.75	16111.75
Normal force (N)	[Fnorm]	47107.60	47107.60
Nominal circumferential force per mm (N/mm)	[w]	570.25	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	42480.31	
Axial force (N)	[Faw]	8446.48	8446.48
Radial force (N)	[Frw]	18525.82	
Circumferential speed reference circle (m/s)	[v]	5.15	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	5.27	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.7
Running-in value (µm)	[yf]	0.8
Correction factor	[CM]	0.800
Gear blank factor	[CR]	1.000
Basic rack factor	[CB]	0.913
Material coefficient	[E/Est]	1.000
Single stiffness (N/mm/µm)	[c']	13.538
Theoretical single stiffness (N/mm/µm)	[c'th]	18.872
Contact stiffness (N/mm/µm)	[cγα]	17.462
Contact stiffness (N/mm/µm)	[cγβ]	14.843
Reduced mass (kg/mm)	[mRed]	0.01839
Resonance speed (min-1)	[nE1]	19619
Resonance ratio (-)	[N]	0.073
Subcritical range		

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.8
Bearing distance l of pinion shaft (mm)	[l]	331.724
Distance s of pinion shaft (mm)	[s]	67.560
Outside diameter of pinion shaft (mm)	[dsh]	63.500
Load in accordance with Figure 13, ISO 6336-1:2006	[-]	1

0:a), 1:b), 2:c), 3:d), 4:e)			
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']		-0.80
Without stiffening			
Effective flank line deviation (µm)	[Fβy]	4.67	
from deformation of shaft (µm)	[fsh·B1]	7.89	
fsh (µm)	[fsh]	78.91	
B1	[B1]	0.10	
fHβ5 (µm)	[fHβ5]	8.00	
Flank line:	crowned with helix angle modification		
Factors from ISO 9085:2002			
Position of contact pattern:	favorable		
from production tolerances (µm)	[fmα*B2]	7.78	
B_2=0.50			
Tooth trace deviation, theoretical (µm)	[Fβx]	5.50	
Running-in value (µm)	[yβ]	0.82	

5.4 K factors

Dynamic factor	[Kv]	1.018
Face load factors		
- Flank	[KHβ]	1.060
- Tooth root	[KFβ]	1.051
- Scuffing	[KBβ]	1.060
Transverse load factors		
- Flank	[KHα]	1.025
- Tooth root	[KFα]	1.025
- Scuffing	[KBα]	1.025
Application factor	[KA]	1.000
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1 ----------	Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.4866	0.4069
Tooth form factor	[YF]	1.252	1.279
Stress correction factor	[YS]	1.979	2.035
Load application angle (°)	[αFen]	26.714	22.738
Load distribution influence factor	[fε]	0.835	
Load application diameter, virtual spur gear (mm)	[d_{en}]	77.964	306.338
Load application diameter (mm)	[d_{en}]	75.680	296.744
Bending moment arm (mm)	[hFe]	5.483	6.375
(-)	[hFe*]	1.218	1.417
Tooth root thickness at critical cross section (mm)	[sFn]	9.691	10.503
(-)	[sFn*]	2.153	2.334
Root fillet radius at critical cross section (mm)	[ρF]	2.143	2.034
(-)	[ρF*]	0.476	0.452
Tangent contact point, virtual spur gear (mm)	[x,y]	4.845 /30.746	
Tangent contact point, virtual spur gear (mm)	[x,y]		5.252 /140.845
Diameter at critical cross section (mm)	[d_{sFn}]	62.250	281.887
Tangent at critical cross section (°)	[$α_{sFn}$]	30.000	30.000
Notch parameter	[q_s]	2.262	2.582
Helix angle factor	[Yβ]	0.960	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00

Effective facewidth (mm)	[beff]	78.74	76.20
Nominal stress at tooth root (N/mm²)	[σF0]	291.74	316.86
Tooth root stress (N/mm²)	[σF]	320.01	347.57
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	0.998	1.001
Surface factor	[YRrelT]	0.964	0.964
Size factor, tooth root	[YX]	1.000	1.000
Life factor	[YNT]	1.000	1.000
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.962	0.965
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	1000.00	1000.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	686.91	689.02
Limit strength tooth root (N/mm²)	[σFG]	961.67	964.63

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	3.01	2.78
Transmittable power (kW)	[kWRating]	480.23	443.51

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.265	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.849	
Helix angle factor	[Zβ]	1.009	
Effective facewidth (mm)	[beff]	76.20	
Nominal contact stress (N/mm²)	[σH0]	1181.05	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1241.86	
Coefficient [fZCa] 1.07 (Helical gear sets with suitable flank modifications acc. to manufacturers experience)			
Single tooth contact factor	[ZB,ZD]	1.03	1.03
Contact stress (N/mm²)	[σHB, σHD]	1284.59	1284.59
Lubricant coefficient for N_L	[ZL]	1.020	1.020
Speed factor at N_L	[ZV]	0.983	0.983
Roughness factor for N_L	[ZR]	1.025	1.025
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	1.000	1.000
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		1.028	1.028
Limited pitting is permissible: No			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1541.29	1541.29
Pitting stress limit (N/mm²)	[σHG]	1541.29	1541.29

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.24	1.24
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.20	1.20
Safety regarding transmittable torque	[SHBD²]	1.44	1.44
Transmittable power (kW)	[kWRating]	322.07	322.07

8 Micropitting

Calculation method according to		ISO/TS 6336-22:2018
Lubricant load according to FVA Info sheet 54/7 10, TR30 example 8: nu40 input; other oil properties are ancillary defaults		
Reference data FZG-C Test:		
Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	127.666
Contact temperature (°C)	[θBRef,A]	251.735
Lubrication gap thickness (µm)	[hRef,A]	0.051
Specific film thickness in test	[λGFT]	0.102

		------- Gear 1	---------- Gear 2 -------
Calculation of permissible specific film thickness			
Material coefficient	[WW]	1.000	
Permissible specific film thickness	[λGFP]	0.143	

Interim results in accordance with ISO/TS 6336-22:2018			
Coefficient of friction	[μ_m]	0.062	
Lubricant factor	[XL]	1.000	
Roughness factor	[XR]	0.937	
Lubrication coefficient for lubrication type	[XS]	1.000	
Tooth mass temperature (°C)	[θM]	88.789	
Tip relief factor	[XCa(A)]	1.322	
Loss factor	[HV]	0.164	
Equivalent Young's modulus (N/mm²)	[Er]	226373.626	
Pressure-viscosity coefficient (m²/N)	[α38]	0.02162	
Dynamic viscosity (Ns/m²)	[ηtM]	22.257	
Roughness average value (µm)	[Ra]	0.380	

Calculation of speeds, load distribution and flank curvature according to method B following ISO/TS 6336-22:2018.

C_a taken as optimal in the calculation. 0=no, 1=yes		0	0
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)		0	
Diameter (mm)	[dy]	65.878	301.550
Relative radius of curvature (mm)	[ρred]	6.240	
Load sharing factor	[XY]	0.713	
X_Y interpolated between the values $X_Y(\varepsilon_b=0.8)$ and $X_Y(\varepsilon_b=1.2)$ according to ISO/TC60/WG6.			
Contact stress (N/mm²)	[pH]	1582.154	
Contact stress (N/mm²)	[pdyn]	1663.621	
Minimal specific film thickness	[λGFY]	0.266	(h_Y=0.101 µm)
Safety against micropitting	[Sλ(B)]	1.864	

9 Safety of hardened layer (subsurface fatigue)

CHD according to		DNV-CG-0036:2021	
		------- Gear 1	---------- Gear 2 -------
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	0.50 / 0.60
Hardening depth CHD using pre-machining (mm)	[t550]	0.72 / 0.82	0.72 / 0.82
Required hardness at t550 (HV)	[HV.t]	550	550
Required hardening depth CHD (mm)	[t400]	0.90 / 1.10	0.90 / 1.10
Required hardness at t400 (HV)	[HV.t]	400	400
Required hardening depth CHD (mm)	[t300]	1.30 / 1.60	1.30 / 1.60
Required hardness at t300 (HV)	[HV.t]	300	300
Condition for hardness on the surface (HV)	[> HV.0]	497	497
Condition for hardness on the surface (HRC)	[> HRC.0]	49	49
HV.0 is calculated with t_z/a_H = 0.5, S_H = min(S_{EHT}, S_{Hssmin})			

Safety of the hardened layer	[SEHT]	1.38	1.38
Required safety	[SHssmin]	1.00	1.00
Result		2	2

-1:Not checked, 0:CHD too shallow, 1:CHD too deep or HV.0 too large, 2:Check performed,
3:Check performed with insufficient hardening depths (t_z/a_H<0.5, t_z/a_{Hst}<0.6)

10 Tooth flank fracture

Calculation method according to ISO/TS 6336-4 (2019)

Calculation according to Method B

		------- Gear 1 ----------	Gear 2 -------
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)		2	2
Diameter (mm)	[dy]	68.784	297.105
Relative radius of curvature (mm)	[ρred]	10.184	10.184
Load sharing factor	[XCP]	0.841	0.841
Contact stress (N/mm²)	[pH]	1345.008	1345.008
Contact stress (N/mm²)	[pdyn]	1414.264	1414.264
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	
Required hardening depth CHD (mm)	[t550]		0.50 / 0.60
The calculation will be performed with the minimum hardening depth			
Core hardness (HV)	[HV_{core}]	302	302

Hardness curve from:

HV(y) gear 1	ISO/TS 6336-4, chapter 7.3.3, Method C1	
HV(y) gear 2	ISO/TS 6336-4, chapter 7.3.3, Method C1	
Maximum material exposure	[AFFmax]	0.745 / 0.745
Depth of maximum load (mm)	[y]	-1.100 / -1.100
Material coefficient	[$K_{material}$]	1.131 / 1.131
Material shearing strength (N/mm²)	[τper]	143.383 / 143.383
Equivalent stress state (N/mm²)	[τeff]	101.139 / 101.139

Safety factor according to ISO DTS 19042-1	[SFF]	1.258	1.258

($S_{FF} = 1/(A_{FF,max} + 0.2) + 0.2$)

11 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1 ----------	Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
Relevant tip relief (µm)	[Ca]	13.00	13.00
Optimal tip relief (µm)	[Ceff]	32.66	
C_a taken as optimal in the calculation. 0=no, 1=yes		0	0
		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.207	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	14
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	76.200	
Applicable circumferential force/facewidth (N/mm)	[wBt]	630.492	

$K_{Bγ} = 1.207$, $w_{Bt} \cdot K_{Bγ} = 761.093$

Angle factor	[Xαβ]	1.025

$\varepsilon_1 = 0.846$, $\varepsilon_2 = 0.540$

11.1 Flash temperature-criteria

		Pair	
Lubricant factor	[XL]	0.844	
Tooth mass temperature (°C)	[θMi]	93.84	
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$			
Average flash temperature (Δ°C)	[θflm]	29.45	
Scuffing temperature (°C)	[θS]	451.99	
Γ coordinates (point of highest temperature)	[Γ]	0.813	
[Γ.A]= -0.519, [Γ.E]= 0.813			
Maximum flash temperature (Δ°C)	[θflamax]	64.80	
Maximum contact temperature (°C)	[θBmax]	158.64	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Approach factor	[XJ]	1.000	
Load sharing factor	[XΓ]	0.713	
Dynamic viscosity (mPa*s)	[ηM]	30.07	(80.0 °C)
Coefficient of friction	[μ_m]	0.059	
Required safety	[SBmin]	2.000	
Margin of safety for scuffing, flash temperature	[SB]	4.731	

11.2 Integral temperature-criteria

		Pair	
Lubricant factor	[XL]	1.000	
Tooth mass temperature (°C)	[θMC]	93.67	
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$			
Integral scuffing temperature (°C)	[θSint]	455.17	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Running-in factor, well run in	[XE]	1.000	
Contact ratio factor	[Xε]	0.275	
Dynamic viscosity (mPa*s)	[ηOil]	30.07	(80.0 °C)
Mean coefficient of friction	[μ_m]	0.062	
Geometry factor	[XBE]	0.331	
Meshing factor	[XQ]	1.000	
Tip relief factor	[XCa]	1.322	
Mean flash temperature (Δ°C)	[θflaint]	19.53	
Integral tooth flank temperature (°C)	[θint]	122.96	
Required safety	[SSmin]	1.800	
Safety factor for scuffing (intg.-temp.)	[SSint]	3.702	
Safety factor for transmitted moment (int.-T.)	[SSL]	8.734	

12 Measurements for tooth thickness

12.1 Tooth thickness tolerances and inputs

12.1.1 Information on pre-machining

		Gear 1	Gear 2
Min. finishing stock (mm)	[qmin]	0.220	0.220
Tooth thickness tolerance pre-machining (mm)	[TsnV]	0.000	0.000
Additional measure for pre-machining (mm)	[ΔAs_p.e/i]	0.468 / 0.468	0.468 / 0.468
Tooth thickness allowance (normal section) (mm)	[As_p.e/i]	0.289 / 0.289	0.362 / 0.362

12.1.2 Information on final machining

Tooth thickness tolerance		Own Input	Own Input
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.179 / -0.179	-0.106 / -0.106

12.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
12.2.1 Information on pre-machining			
Number of teeth spanned	[k]	3.000	9.000
Base tangent length with allowance (mm)	[Wk.e/i]	36.146 /36.146	118.798 /118.798
(mm)	[ΔWk.e/i]	0.272 / 0.272	0.340 / 0.340
12.2.2 Information on final machining			
Number of teeth spanned	[k]	3.000	9.000
Base tangent length (no backlash) (mm)	[Wk]	35.874	118.458
Base tangent length with allowance (mm)	[Wk.e/i]	35.706 /35.706	118.358 /118.358
(mm)	[ΔWk.e/i]	-0.168 / -0.168	-0.100 / -0.100
Diameter of measuring circle (mm)	[dMWk.m]	73.422	294.765

12.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
12.3.1 Information on pre-machining			
Effective diameter of ball/pin (mm)	[DMeff]	10.000	8.000
Diametral measurement over two balls (mm)	[MdK.e/i]	89.364 /89.364	304.866 /304.866
12.3.2 Information on final machining			
Theoretical diameter of ball/pin (mm)	[DM]	9.606	7.872
Effective diameter of ball/pin (mm)	[DMeff]	10.000	8.000
Radial single-ball measurement, no backlash (mm)	[MrK]	44.665	152.054
Radial single-ball measurement (mm)	[MrK.e/i]	44.518 /44.518	151.929 /151.929
Diameter of measuring circle (mm)	[dMMr.m]	73.783	292.776
Diametral measurement over two balls, no backlash (mm)	[MdK]	88.896	304.017
Diametral measurement over two balls (mm)	[MdK.e/i]	88.602 /88.602	303.766 /303.766
Diametral measurement over pins, no backlash (mm)	[MdR]	89.330	304.109
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	89.035 /89.035	303.858 /303.858
Measurement over 2 pins, free, AGMA 2002 (mm)	[dk2f.e/i]	88.580 /88.580	303.762 /303.762
Measurement over 2 pins, transverse, AGMA 2002 (mm)	[dk2t.e/i]	89.441 /89.441	303.949 /303.949
Measurement over 3 pins, axial, AGMA 2002 (mm)	[dk3A.e/i]	89.035 /89.035	303.858 /303.858

12.4 Tooth thickness

		------- Gear 1 ----------	Gear 2 -------
12.4.1 Information on pre-machining			
Medium tip diameter (mm)	[da.m]	82.190	301.550
Reference chordal height from da.m (mm)	[hac]	7.005	6.438
Tooth thickness at height hac, chord (mm)	[sc.e/i]	9.106 / 9.106	8.869 / 8.869
Tooth thickness on reference circle, arc (mm)	[sn.e/i]	9.131 / 9.131	8.870 / 8.870
12.4.2 Information on final machining			
Medium tip diameter (mm)	[da.m]	82.190	301.550
Reference chordal height from da.m (mm)	[hac]	6.987	6.432
Tooth thickness at height hac, chord (mm)	[sc]	8.819	8.506
(mm)	[sc.e/i]	8.646 / 8.646	8.401 / 8.401
Tooth thickness on reference circle, arc (mm)	[sn]	8.842	8.508
(mm)	[sn.e/i]	8.662 / 8.662	8.402 / 8.402

12.5 Backlash

		Pair
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.000 / 0.000
Radial backlash (mm)	[jrw.e/i]	0.343 / 0.343
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.297 / 0.297
Normal backlash (mm)	[jnw.e/i]	0.268 / 0.268
Center distance allowances (mm)	[Aa.e/i]	0.000 / 0.000
Backlash free center distance (mm)	[aControl.e/i]	182.537 /182.537
Backlash free center distance, allowances (mm)	[jta]	-0.343 / -0.343

		Gear 1	Gear 2
dNf.i with aControl (mm)	[dNf0.i]	65.546	286.484
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.164	1.225
Tip clearance (mm)	[c0.i(aControl)]	1.387	1.370

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.1153/ 0.1153	(0°6'55"/ 0°6'55")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.4843/ 0.4843	(0°29'3"/ 0°29'3")

13 Gear tolerances

According to ISO 1328-1:2013, ISO 1328-2:1997		Gear 1 [A]		Gear 2 [A]	
Single pitch tolerance (µm)	[fpT]	± 9.50	(-)	± 10.00	(-)
Base pitch tolerance (µm)	[fpbT]	± 8.91	(-)	± 9.38	(-)
Sector pitch tolerance (µm)	[Fpz/8T]	± 20.00	(6)	± 23.00	(6)
Cumulative pitch tolerance, total (µm)	[FpT]	28.00	(6)	35.00	(6)
Adjacent pitch difference tolerance (µm)	[fuT]	14.00	(6)	14.00	(6)
Profile form tolerance (µm)	[ffαT]	11.00	(-)	11.00	(-)
Profile slope tolerance (µm)	[fHαT]	± 8.50	(6)	± 8.50	(6)
Profile tolerance, total (µm)	[FαT]	13.00	(6)	14.00	(6)
Helix form tolerance (µm)	[ffβT]	12.00	(6)	13.00	(6)
Helix slope tolerance (µm)	[fHβT]	± 11.00	(-)	± 11.00	(-)
Helix tolerance, total (µm)	[FβT]	16.00	(6)	17.00	(6)
Runout tolerance (µm)	[FrT]	25.00	(6)	32.00	(6)
Single flank composite tolerance, total (µm)	[FisT]	38.00	(6)	45.00	(6)
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	9.50	(6)	9.50	(6)
Radial composite tolerance, total (µm)	[Fi"]	44.00	(6)	60.00	(6)
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	22.00	(6)	22.00	(6)

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020		[R]		[R]	
Radial composite tolerance, total (µm)	[FidT]	35.00	(40)	52.00	(41)
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	18.00	(40)	22.00	(41)

Own input for tolerance: (-)

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99		
Total helix deviation for calculation (µm)	[Fβ]	17.00
Distance between bearings for calculation (mm)	[LG]	331.72
Maximum value for axis deviation error (µm)	[fΣβ]	± 37.00
Maximum value for axis inclination error (µm)	[fΣδ]	± 74.01

14 Modifications and determination of the tooth form

14.1 Profile and flank line modifications

Gear 1

Symmetric (both flanks)

- Tip relief, linear

Caa = 13.000 µm LCa = 0.500 ·mn dCa = 79.478 mm

Gear 2

Symmetric (both flanks)

- Tip relief, linear

Caa = 13.000 µm LCa = 0.500 ·mn dCa = 299.597 mm

14.1.1 Tip relief verification

		------- Gear 1 ----------	Gear 2 -------
Diameter (mm)	[dcheck]	82.100	301.460
Tip relief left/right (µm)	[Ca L/R]	12.6 /12.6	12.4 /12.4
Diameter (mm)	[dNa.i]	82.190	301.550
Tip relief left/right (µm)	[Ca L/R]	13.0 /13.0	13.0 /13.0

14.2 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

15 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	185.028

15.1 Masses, stiffnesses and moments of inertia

		------- Gear 1 ----------	Gear 2 -------
Mass (kg)	[m]	2.492	39.675
Total mass (kg)	[Σm]	42.167	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.00160	0.41990
System (da+df)/2...di (kg*m²)	[J]	0.02541	

15.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	1.149
Torsion when subjected to nominal torque (°)	[δcr]	0.074
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	87.277

15.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.059
Wear sliding coefficient by Niemann	[ζw]	0.779
Loss factor	[HV]	0.164
Gear power loss (kW)	[PVZ]	2.171
Meshing efficiency (%)	[ηz]	99.030

16 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	10000.0000
Service life (calculated with required safeties):		
System service life (h)	[H_{att}]	> 1000000

		------- Gear 1 ----------	Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H_{Hatt}]	1e+06	1e+06

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

16.1 Damage

Damage relative to the required service life (H, 10000.0 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	0.0000	0.0000	0.0000

17 Reliability calculation

17.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

17.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	<TD CENTER>β	<TD CENTER>t_0	<TD CENTER>T	<TD CENTER>R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	85800	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	85800	1.300e+00	9.014e+29	4.295e+30	100.0000
2	Tooth root	20429	1.700e+00	9.654e+29	1.484e+30	100.0000
2	Tooth flank	20429	1.300e+00	9.014e+29	4.295e+30	100.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

17.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	10000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	100.0000
Reliability R, gears subsystem (%)	[R_{subG}]	100.0000
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	> 1'000'000
Service life H, tooth flanks subsystem (h)	[H_{subH}]	> 1'000'000
Service life H, gears subsystem (h)	[H_{subG}]	> 1'000'000

18 Remarks

18.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

18.2 Calculations and factors

- Calculation of Z_{β} acc. to Corrigendum 1 ISO 6336-2:2008 with Z_{β} = 1/(COS(β)^0.5)

- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

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